Week 4 — Session 2

Topic: Competitive Problem Patterns

Language: C++17

Focus: direct comparison, hashing, and recognizing common competitive-programming patterns.

Task 1: Compare the Triplets

Practice: https://www.hackerrank.com/challenges/compare-the-triplets/problem

Compare Alice's and Bob's three scores position by position and award one point to the higher score.

Solution Code

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    vector<int> a(3), b(3);
    for (int &x : a) cin >> x;
    for (int &x : b) cin >> x;

    int alice = 0, bob = 0;

    for (int i = 0; i < 3; ++i) {
        if (a[i] > b[i]) alice++;
        else if (a[i] < b[i]) bob++;
    }

    cout << alice << ' ' << bob << '\n';
    return 0;
}
```

Task 2: Contains Duplicate

Practice: https://leetcode.com/problems/contains-duplicate/

Determine whether any value occurs at least twice in the array.

Solution Code

```
#include <bits/stdc++.h>
using namespace std;

bool containsDuplicate(const vector<int>& nums) {
    unordered_set<int> seen;

    for (int x : nums) {
        if (seen.count(x))
            return true;
        seen.insert(x);
    }

    return false;
}

int main() {
    int n;
    cin >> n;

    vector<int> nums(n);
    for (int &x : nums) cin >> x;

    cout << (containsDuplicate(nums) ? "true" : "false") << '\n';
    return 0;
}
```

All programs are standalone C++17 solutions and can be compiled with a standard C++17 compiler.